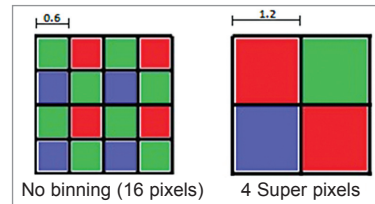


When doing a 2×2 binning, the size of the pixel is doubled whereas the effective resolution gets divided by 4.

For example, a binned image from a 16MP sensor with pixels measuring $0.6\mu\text{m}$ would only output a 4MP image, but the size of each pixel would double to $1.2\mu\text{m}$. Fig. 4 shows the result of 2×2 pixel binning on a segment containing sixteen pixels.



element most commonly known as a pixel is the smallest addressable element of any picture. Pixels are uniformly arranged in a 2D grid to form a picture. The total count of pixels is denoted by megapixels, while the number of pixels in the horizontal and vertical axes determine the resolution of an image.

Resolution in photography means spatial separation and determines the quality of an image. The camera resolution is specified simply by multiplying the pixels available in horizontal and vertical lines within a unit area. Image sensors with higher resolution (more pixels) enable

pictures with greater detail and therefore enlargement.

A general rule of thumb is that the sensors with smaller pixels require higher resolution than sensors with larger pixels to achieve the same quality of image. Fig. 3 shows how different resolutions can be compared with each other for similar sized pixels.

Pixel binning. Pixel bin-

ning is a technique used to improve the low lighting photography of a portable camera. The quality of image depends on the amount of light captured by the pixels, and larger pixels are better at capturing more light due to their greater surface area. The pixel size is measured in microns.

The latest smartphone cameras are sporting extremely high resolution but with smaller pixel width. The higher resolution works well in bright lighting but sadly not so well in poor lighting conditions. That's where pixel binning comes into play. Pixel binning is a technique where multiple pixels are combined to form a single larger pixel.

As shown in Fig. 4, sixteen pixels undergo 2×2 binning to create four super pixels. These super pixels will have better balance between brightness and colours.

Devices with very high resolution prefer to use pixel binning. Some smartphones using pixel binning are Redmi Note 10, Google Pixel 6 series, Samsung Galaxy S21, and Xiaomi Mi11.

Monochrome and colour sensor.

Cameras can detect a wide range of electromagnetic waves. Cameras can be classified into 2 categories, a colour camera and a monochrome camera.

Colour cameras are capable of distinguishing different shades of colours, whereas monochrome cameras can only produce output in different shades of a single colour. As shown in Fig. 5, the colour sen-

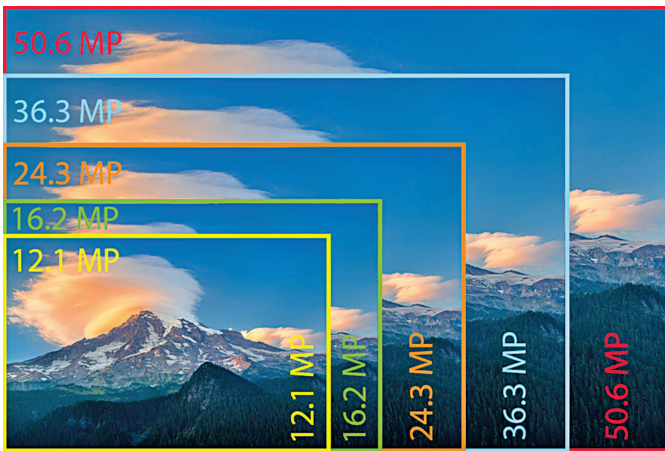


Fig. 3: Comparison of different resolutions with same pixel size (Source: Link)

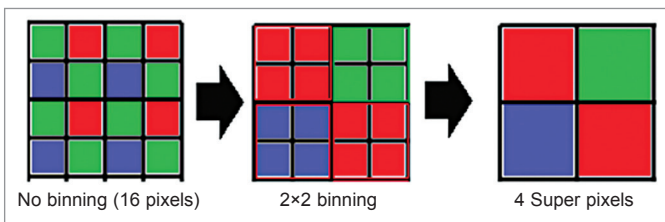


Fig. 4: How 2×2 pixel binning works

TABLE 2
COMPARISON OF DIGITAL, MACHINE-VISION, AND SURVEILLANCE CAMERAS

Machine vision	Surveillance camera	Digital cameras
Need for monitoring production line	Need for monitoring people	Mostly used for clicking casual pictures and videos
Sends uncompressed data for processing	Sends compressed data for processing	Sends compressed data for processing
Can be employed with casing	Needs to be extremely robust	Needs to be small and power-saving
Less tolerant to poor lighting	Tolerant to poor lighting	Able to work in various lighting conditions
Used in a closed protected environment	Mostly used in open unprotected environment	Mostly used in open unprotected environment
For example, used for quality control in factory automation	For example, used for intelligent traffic monitoring system, such as road tolling	For example, used for casual photography and videography